RESEARCH / SOLUTION ARCHITECTURE

Market Basket Analysis & Collaborative Filtering

Finds what Bronze-tier customers buy together and recommends the next product.

BUILT WITH

- Python
- Recommendation Systems
- Market Basket Analysis
- Pandas
- Scikit-learn
- Jupyter Notebook

806K+
transactions analysed

2
independent segmentations: RFM and K-Means

0
rules at the strict threshold, reported not hidden

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How it works

1 Transactions

Online Retail II, 806K+ rows, cleaned of cancelled and missing-ID orders.

2 RFM segments

Recency, frequency and monetary quartiles give Gold, Silver and Bronze.

3 K-Means check

A second, independent segmentation (k=4) cross-checks the tiers.

4 Apriori on Bronze

A strict threshold (lift 6 or more) gave an empty result, reported as a finding.

5 Collaborative filtering

Co-occurrence and Pearson correlation give per-customer recommendations.

Key decisions

01

Two segmentations, not one

RFM quartiles and K-Means clustering ran on the same features. Getting a similar structure from two methods is a basic sanity check.

02

Scoped to one segment

Bronze customers (low frequency, low spend, still active) have the most upside, so recommendations were built there.

03

Report the empty result

At lift 6 or more and confidence 0.70 or more, no rules qualified. The threshold was not loosened until something appeared.

04

Name the weak spots

Exact correlations of 1.0 or -0.5 on thin data are a known artefact, and the write-up says so.

Stack, status and what is next

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KNOWN LIMITS

- Correlations from thin evidence can look more confident than they are.

NEXT STEPS

- Re-run Apriori at a lower threshold to see weaker real associations.
- Weight correlations by the number of shared invoices.
- Repeat for the Silver and Gold segments.

SOURCE https://github.com/walethewave/Advanced-Retail-Analytics-Suite-Customer-Segmentation-Strategic-Insights